

Institution
Address

July 6, 2026

Program Chairs
CVPR 2026 EarthVision Workshop

Dear Program Chairs,

This file is retained as a workshop-format cover-letter template for Paper 12. The active journal submission package is maintained under `submission/paper12_isprs_jprs_20260606`; the wording below is synchronized with the current bounded manuscript rather than the earlier public-benchmark-only draft.

We are pleased to submit our manuscript, “Architecture-Aware Diagnosis of Parameter-Efficient Adaptation in Prithvi-100M: Fused-QKV Failure, Capacity Boundaries, and Cross-Domain Remote-Sensing Validation,” for consideration at the CVPR 2026 EarthVision Workshop.

Summary. The paper studies how parameter-efficient fine-tuning (PEFT) methods behave on Prithvi-100M when sensor inputs, channel bridges, deployment labels, and transfer domains differ from pre-training. We evaluate seven adaptation strategies across EuroSAT, BigEarthNet-S2, LandCover.ai, a production-style Linhe County LULC workflow supervised by Esri-derived labels, and LoveDA urban/rural cross-domain transfer. The current manuscript also includes a completed EuroSAT capacity sweep, a channel-bridge ablation, and a Linhe synthetic weak-label control that is not an independent manual validation set. Three bounded findings emerge:

1. Standard module-wise LoRA is vulnerable to a fused-QKV insertion trap on Prithvi-100M, while corrected split-QKV LoRA still remains below Housby adapters in the tested Prithvi-100M setting.
2. Housby adapters provide the most reliable PEFT baseline in our experiments, but the result is scoped to one backbone and should not be read as a universal GeoFM ranking.
3. Domain shift, label source, channel bridging, and adapter capacity materially affect the operating point, with LoveDA supporting an adapter-capacity hypothesis rather than a general rule for all geospatial foundation models.

Relevance to EarthVision. The paper addresses a practical gap in geospatial AI: practitioners often transfer PEFT intuitions from NLP and conventional vision without checking whether remote-sensing backbones expose the same trainable tensors or tolerate the same adaptation modules. Our benchmark and audit files make that inspection reproducible for Prithvi-100M, including the fused-attention insertion check, channel-bridge sensitivity, trainable-parameter accounting, and per-class cross-domain diagnostics.

Novelty and contribution. The contribution is diagnostic rather than universal. We separate implementation failure from post-fix low-rank underperformance, show that input-stage adaptation is regime-dependent, and demonstrate that deployment-style labels and geographic scene splits can change the apparent PEFT delta. The result is a stress-test protocol for Prithvi-like GeoFM adaptation rather than a claim that any method replaces a full GIS production stack.

Confirmation. This manuscript has not been published or submitted elsewhere in the form described here.

All authors have approved the submission.

Sincerely,

[Author Name]